9 The equation of a curve is $y = 24\sqrt{x} - 8x^{\frac{3}{2}} + 16$.

(a) Find $\frac{dy}{dx}$. **[3]**

(b) Find the coordinates of the turning point. **[3]**

(c) Determine the nature of the turning point. **[2]**

9(a)

9(b)

9(c)